

E-COMMERCE SALES AND CUSTOMER ANALYSIS **USING PYTHON**

**RETAIL TRANSACTIONS TRACKING CUSTOMER DEMOGRAPHICS,
PRODUCT PERFORMANCE**



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Executive Summary:

This report presents an analysis of e-commerce sales performance based on 5,000 customer transaction records. The analysis examines revenue, product categories, regional performance, payment methods, delivery times, and customer ratings. The findings indicate differences in revenue across product categories and regions, while customer ratings and delivery performance also show variation. Based on the findings, areas with relatively lower performance should be examined further to identify potential opportunities for improvement.

INTRODUCTION:

E-commerce businesses generate large amounts of Transactional and customer-related data. Analyzing this information can help organizations understand sales performance, customer behavior, operational efficiency, and areas requiring improvement. This project uses Python to analyze an e-commerce dataset and identify meaningful patterns across revenue, customers, products, regions, payment methods, delivery time, discounts, and customer ratings.

OBJECTIVES:

- Analyze revenue
- Identify the best categories
- Analyze regional revenue
- Identify top customers
- Examine payment methods
- Identify best delivery days
- Discount analysis
- Explore the relationship between discounts and revenue

Data Source:

The dataset was obtained from Kaggle and contains 5,000 unique records and 12 analytical columns. It includes demographic, categorical, logistical, and financial information related to e-commerce sales.

Data Preparation:

Missing values and duplicate records were examined; however, no missing values or duplicate records were identified. The order_date column was converted to the datetime format and prepared for further analysis. Additional time-based features including year, month, year_month and month_name were created.

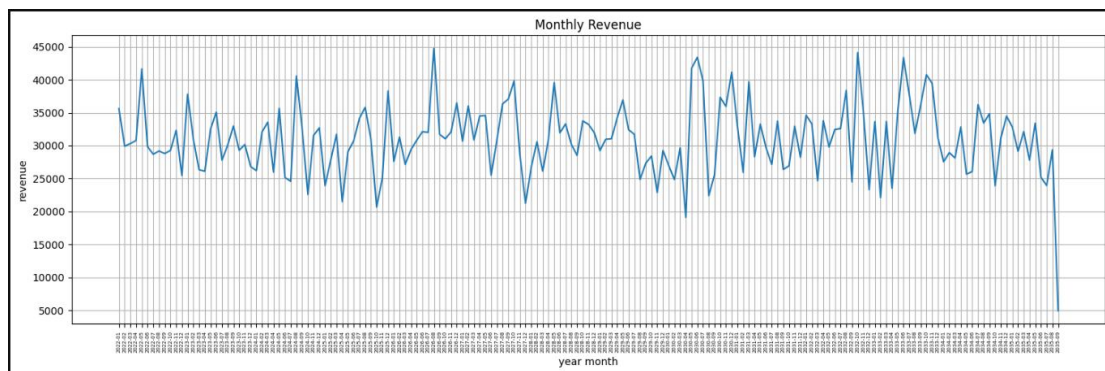
Exploratory Data Analysis

Overall Sales Performance:

- Total Orders : 5000
- Total Customers : 989
- Total Revenue : Rs. 5,109,775.74
- Total Quantity : 20224
- Average Order Value : 1021.9551
- Average Customer Rating : 2.9740
- Average Delivery Time : 6.1188 Days

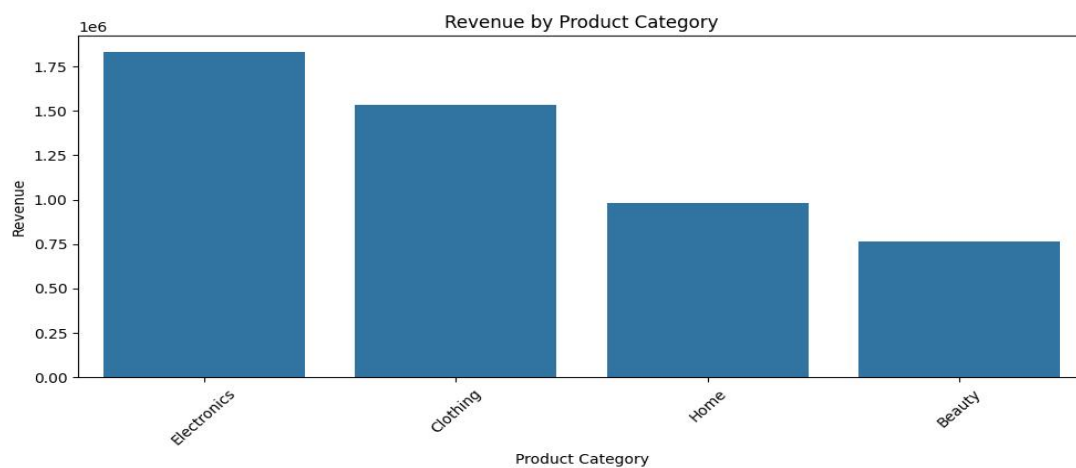
The dataset contains 5000 orders from 989 customers, generating total revenue 5109775.74 Rs. The average order value was 1021.9551 and average delivery time was 6.1188 days.

Monthly Revenue:



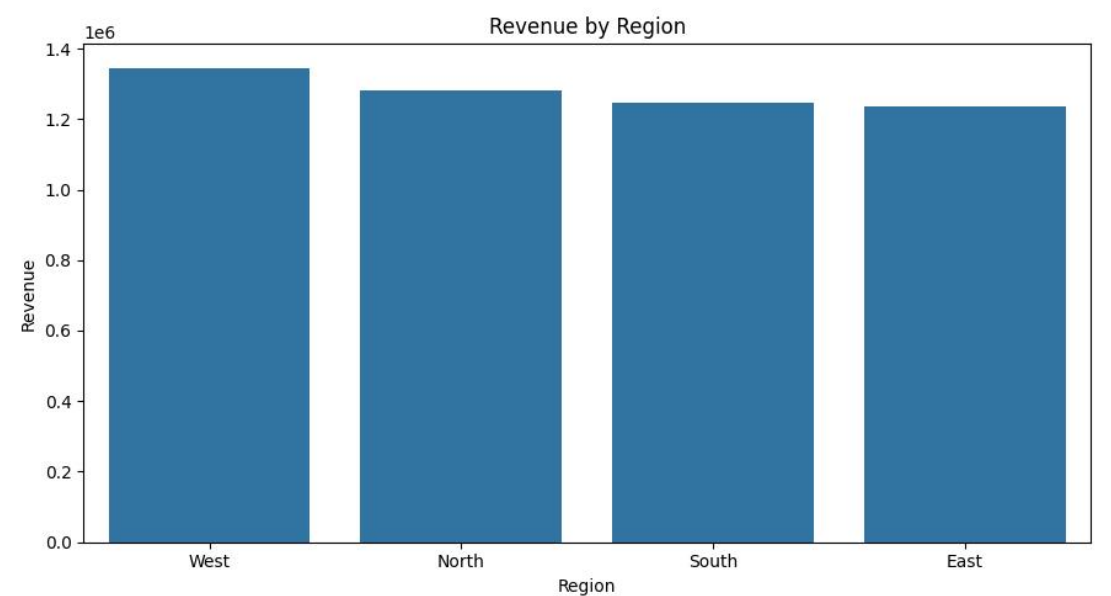
Monthly revenue analysis shows variation in sales across the analyzed period. Revenue reached its highest level in 2026-08, while the lowest revenue was recorded in 2030-03.

Product Category Analysis:



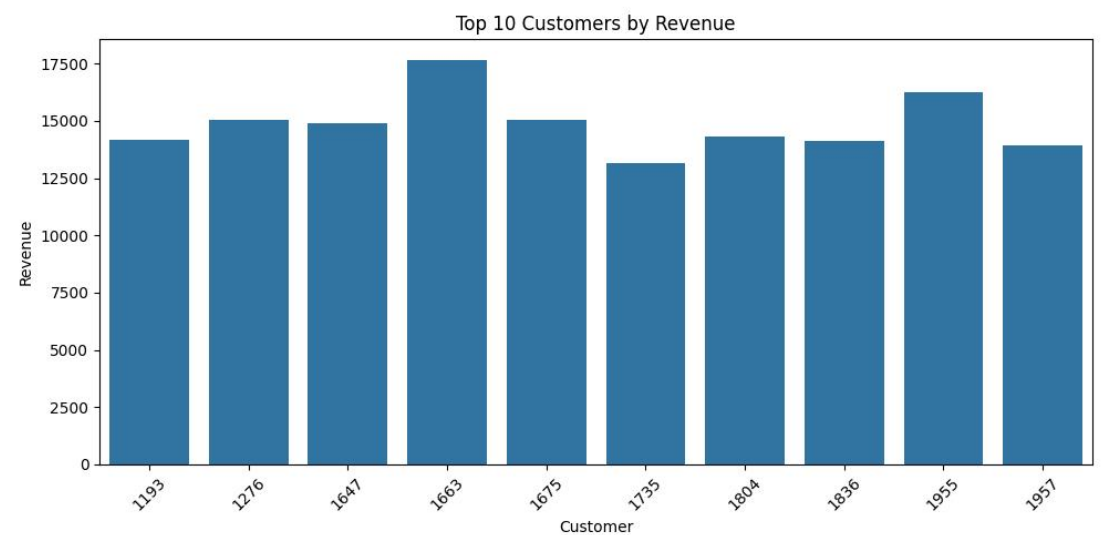
The analysis shows the revenue generated across different product categories. Electronics generated the highest revenue, amounting to Rs. 1,829,899.22, while Beauty generated the lowest revenue, amounting to Rs. 765,860.88.

Regional Analysis:



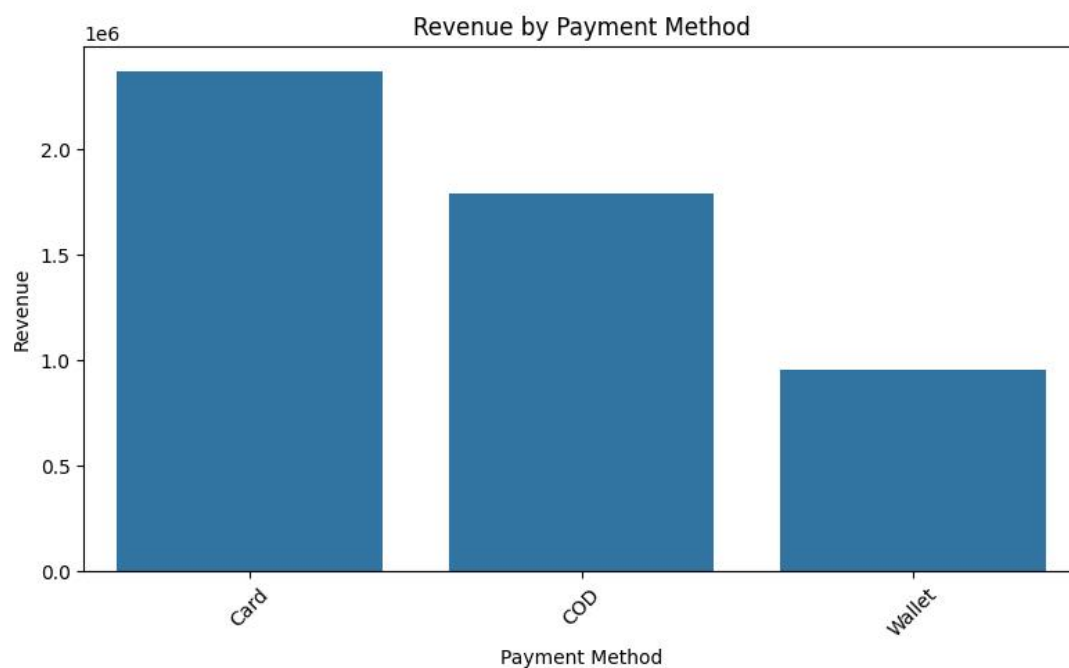
The analysis indicates that the West Region generated the highest revenue, amounting to Rs. 1,345,582.16, while the East Region generated the lowest revenue, amounting to Rs. 1,236,044.23. The findings also show that the West Region recorded more orders than the other regions.

Customer Analysis:



The analysis identifies the top ten customers who generated the highest revenue. Among them, Customer 1663 generated the highest revenue, amounting to Rs. 17,680.69. The number of orders placed by these top ten customers ranged from 7 to 14.

Payment Method Analysis:



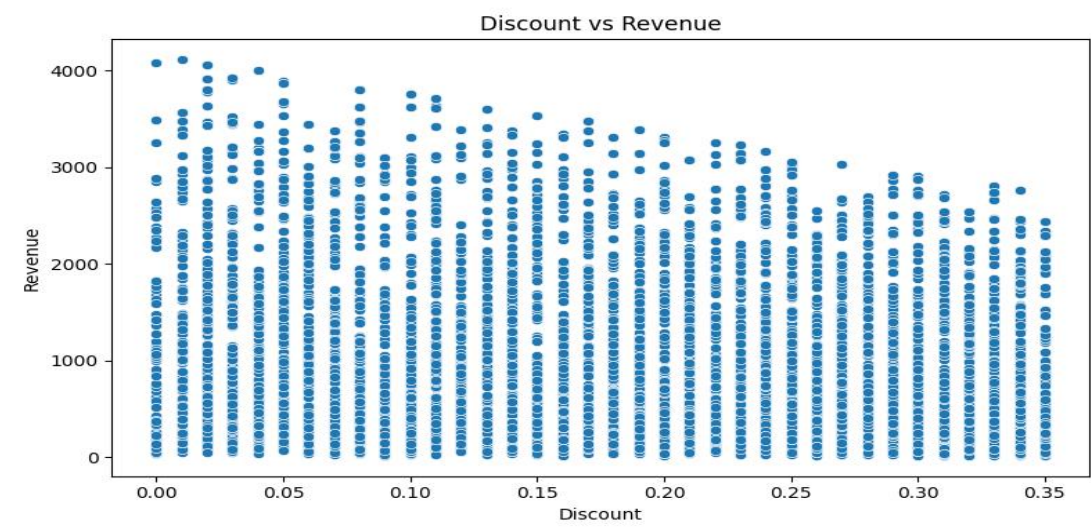
The payment method analysis shows the contribution of each payment method to total revenue. The Card payment method recorded the highest revenue, amounting to Rs. 2,366,248.20, while the Wallet payment method generated the lowest revenue, amounting to Rs. 955,119.10.

Delivery Performance:



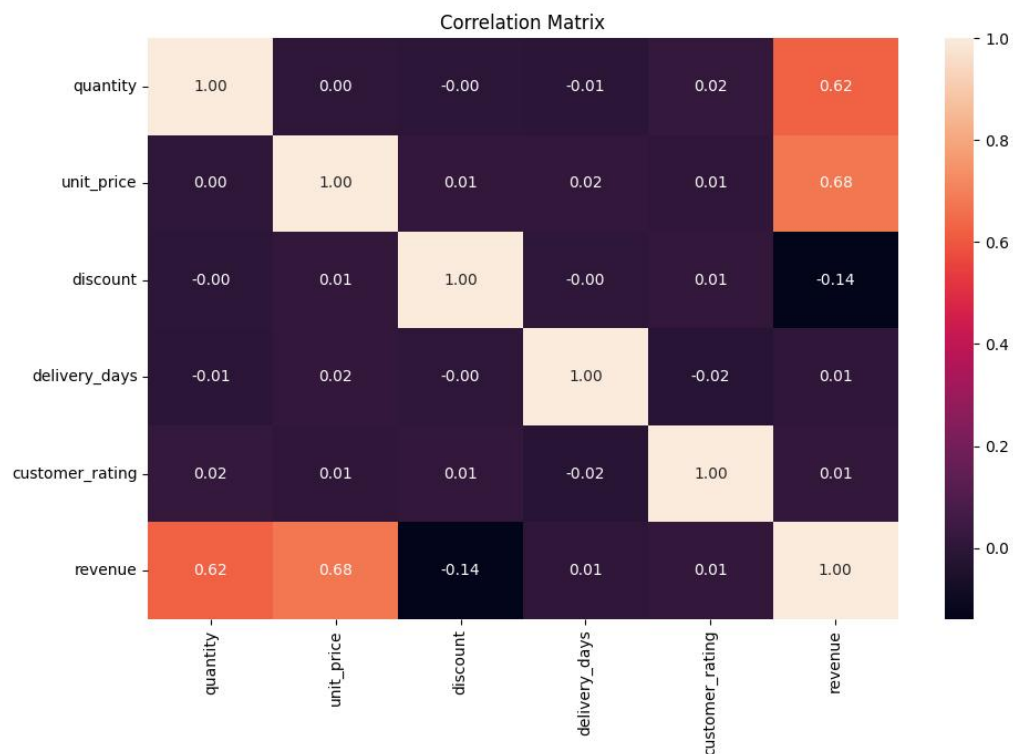
Delivery performance was analyzed alongside customer ratings to determine whether delivery time had an impact on customer satisfaction. The analysis indicates that delivery time had relatively little impact on customer ratings.

Discount Analysis:



The discount analysis indicates that discount levels had a limited impact on revenue performance. The lowest discount level was associated with the highest revenue, suggesting that higher discounts did not necessarily result in higher revenue.

Correlation Analysis:



The correlation analysis shows that revenue is moderately correlated with product quantity. In contrast, revenue is weakly correlated with discount levels.

Key Findings:

- Total revenue was Rs. 5,109,775.74 across 5000 orders.
- Electronics generated the highest category-level revenue.
- West recorded the highest regional revenue.
- Customer 1663 contributed significantly to total revenue.
- Card was the most frequently used/highest-revenue payment method.
- Average delivery time was 6.1188 Days.
- Average customer rating was 2.9740.
- Discount and revenue showed weak relationship.

Business Recommendations:

Based on the findings, management should conduct a detailed review of customer demand, pricing, and distribution performance in the East Region to identify the factors contributing to its lower revenue. The organization should also focus on enhancing the performance of its Electronics category, which recorded the highest revenue.

Furthermore, management should review the current discount levels, as the analysis indicates that discounts had a limited relationship with revenue. The organization should also optimize the Wallet and Cash on Delivery (COD) payment methods to improve payment efficiency and customer convenience.

Conclusion:

This project used Python to analyze e-commerce sales and customer data across multiple dimensions, including revenue, product categories, regions, customers, payment methods, delivery performance, discounts, and customer ratings. The analysis demonstrated how data cleaning, exploratory analysis, statistical summaries, and visualization can be used to identify meaningful business patterns. The resulting findings provide a structured view of sales and customer performance and demonstrate the application of Python-based data analysis to a practical business dataset.